

Appendix A. Derivation of expected payoff functions

This appendix details the derivation of expected payoff functions operationalizing key information governance constructs: verification capacity (u_1, u_2), accountability allocation, and downstream feedback incentives. The variables α , β , and γ represent the probabilities of proactive governance (platform), compliant creation (creator), and reporting (user), respectively.

Appendix A.1. Creator's expected payoffs

Creators trade off compliant creation (stable revenue) against illicit creation (high-return but sanction-sensitive). This formulation embeds platform verification capacity and sanctioning intensity, capturing how governance reshapes violation attractiveness.

Let E_{11} and E_{12} denote expected payoffs for compliant creation (β) and illicit creation ($1 - \beta$). Compliant creation yields E_m regardless of other strategies.

$$E_{11} = E_m \tag{A.1}$$

The payoff for illicit creation is the weighted average of corresponding matrix payoffs:

$$\begin{aligned} E_{12} = & \alpha\gamma[(1 - u_1)(1 - u_2)E_u - C_b - f_{b1} - u_2f_{b2} - R_b] \\ & + \alpha(1 - \gamma)[(1 - u_1)(1 - u_2)E_u - C_b - u_2f_{b1} - u_2f_{b2}] \\ & + (1 - \alpha)\gamma[(1 - u_1)E_u - C_b - f_{b1} - R_b] \\ & + (1 - \alpha)(1 - \gamma)[(1 - u_1)E_u - C_b] \end{aligned}$$

In this formulation, u_1 and u_2 appear multiplicatively, reflecting layered verification: upstream identification (u_1) and in-stream audit (u_2) jointly reduce monetizable exposure probabilities. Simplifying yields Equation (2).

Appendix A.2. Platform's expected payoffs

Platform payoffs encode a core dilemma: proactive verification imposes fixed costs, while passivity risks reputational and regulatory exposure. Let E_{21} and E_{22} denote payoffs for proactive governance (α) and passive governance ($1 - \alpha$).

$$\begin{aligned} E_{21} = & \beta\gamma(-C_{p1} - C_{p2}) + \beta(1 - \gamma)(-C_{p1} - C_{p2}) \\ & + (1 - \beta)\gamma(-C_{p1} - C_{p2} - R_{p1}) + (1 - \beta)(1 - \gamma)(-C_{p1} - C_{p2} - R_{p1}) \end{aligned}$$

Simplifying yields:

$$E_{21} = -C_{p1} - C_{p2} - (1 - \beta)R_{p1} \tag{A.2}$$

Similarly for E_{22} :

$$\begin{aligned} E_{22} = & \beta\gamma(-C_{p1} - f_p) + \beta(1 - \gamma)(-C_{p1} - f_p) \\ & + (1 - \beta)\gamma(-C_{p1} - R_{p1} - R_{p2} - f_p) + (1 - \beta)(1 - \gamma)(-C_{p1} - R_{p1} - f_p) \end{aligned}$$

Here, f_p represents institutional accountability pressure increasing the cost of passivity, while R_{p1} and R_{p2} capture reputational losses. Simplifying yields Equation (4).

Appendix A.3. User's expected payoffs

Users provide downstream feedback signals (reporting) at a time and attention cost. The payoff structure illustrates why user reporting functions as a terminal mechanism: it increases governance signals but does not alter upstream incentives without institutional leverage. Let E_{31} and E_{32} denote payoffs for reporting (γ) and non-reporting ($1 - \gamma$).

$$\begin{aligned} E_{31} = & \alpha\beta(V_m - C_u) + (1 - \alpha)\beta(V_m - C_u) \\ & + \alpha(1 - \beta)[-(1 - u_1)(1 - u_2)V_d - C_u + f_u] \\ & + (1 - \alpha)(1 - \beta)[-(1 - u_1)V_d - C_u + f_u] \end{aligned}$$

Simplifying yields Equation (5).

For E_{32} :

$$\begin{aligned} E_{32} = & \alpha\beta(V_m) + (1 - \alpha)\beta(V_m) \\ & + \alpha(1 - \beta)[-(1 - u_1)(1 - u_2)V_d + u_2f_u] \\ & + (1 - \alpha)(1 - \beta)[-(1 - u_1)V_d] \end{aligned}$$

The term u_2f_u reflects that verification capacity amplifies reporting effectiveness. However, as simulations show, this alone cannot substitute for upstream accountability internalization. Simplifying yields Equation (6).